MILITARY SPECIFICATION
ROCKET MOTOR, XM119
METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers Rocket Motor XM119, Metal Parts Assembly. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following document of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification For
MIL-I-6866	- Inspection, Penetrant, Method of
MIL-I-6868	- Inspection, Process Magnetic Particle

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-413	- Visual Inspection Guide for Rubber "O" Rings

DRAWINGS

Munitions Command

D11740302	- Rocket Motor, XM119, Metal Parts Assembly
F9266824	- Box, Fiberboard, (Inner) Packing, Ammunition, for Rocket Motor XM119
C9266825	- Box, Packing, Ammunition, for Rocket Motor XM119
C9266826	- Box, Fiberboard (Outer) Alternative, Packing Ammunition for Rocket Motor, XM119
LI-11740302	- Inspection Equipment List for Rocket Motor, XM119 Metal Parts Assembly

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(Copies of specifications, standards and drawings required by the suppl in connection with specific procurement functions should be obtained fr procuring activity or as directed by the contracting officer.)

2.2 Other Publications.-The following documents form a part of thi specification to the extent specified herein. Unless otherwise indicat issue in effect on date of invitation for bids or request for proposal apply.

ASA B46.1 - Surface Texture

(Application for copies should be addressed to the American National St Institute (ANSI), 1430 Broadway, New York, N.Y. 10018.)

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the quirements specified on the applicable drawings and referenced specific

3.2 Physical properties.-Physical properties required by the drawi shall be uniform throughout the part within specified tolerances. Requ for all alloy steel parts shall be achieved by quenching and tempering.

3.3 Hydrostatic pressure.-The components listed in Table I shall w stand the corresponding internal pressure for 15 seconds, minimum witho application of any external load and shall show no evidence of le ge mechanical failure. (See 6.2)

TABLE I

Component	Hydrostatic Pressure (psig)
Motor Tube Assembly	7000 to 7500
Cap, End	7000 to 7500
Cap, Head End	7000 to 7500

3.4 Firing pin release.-The firing pin of the firing pin assembly, less safety pin, shall release from the cocked position when the applic of an axial load of 8.5 to 16.5 pounds is applied to the sear.

3.5 Protective finish.-Protective finish shall be in accordance wi applicable specification referenced on the drawings and shall apply to surfaces regardless of configuration.

3.6 Residual magnetism.-The metal part assembly shall not deflect indicator of the compass more than 3 degrees in either direction when p in front of the compass at a distance of 6 inches.

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3.7 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.7.1 Metal defects.-The part shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.7.2 Burr.-The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.7.3 Foreign matter.-The part or assembly shall not contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.2 First article sample

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 6 metal parts assemblies, 10 sets of unassembled components for dimensional and visual examination and tests. The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.-Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

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4.3 Inspection Provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be presented in accordance with MIL-STD-105.

4.3.1.2 Size of lot.-For the purpose of identification and shipment, the shipping lots of metal parts assemblies shall consist of 1200 units maximum, exclusive of test samples, provided the homogeneity of manufacturing process has been retained throughout the inspection lot. A smaller lot may be shipped upon approval of the contracting officer.

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein using the sampling plan and associated criteria in Tables II and III. For an individual defect (single characteristic), the criteria in Table II shall be used. For total defects within a major or minor group, the criteria in Table III shall be used. All nonconforming material shall be rejected. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. The roughness comparison specimens prescribed in ASA B46.1, shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter

TABLE II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 -22000	300	4	5	5	6

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TABLE III

Sample Size for Major Defect No Defect in Class	Total Allowable Defect per Sample Size 110	150	225	300
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over 43	6	8	11	14
Sample Size for Minor Defect No Defect in Class	**110**	**150**	**225**	**300**
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over 48	8	11	17	20

4.3.2.1 Cap, Head End, Drawing C11740304.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Concentricity of groove to large thread	Gage
102. Concentricity of outside diameter to large thread	Gage
103. Concentricity of small thread to large thread	Gage
104. Diameter of groove, maximum	Gage
105. Outer diameter, minimum	Gage
106. Pitch diameter of large thread	Gage
107. Minor diameter of small thread	Gage
108. Pitch diameter of small thread	Gage
109. Minor diameter of small thread	Gage
110. Port chamfer diameter	Gage
111. Port counterbore diameter	Gage
112. Metal defective	Visual
113. Protective finish inadequate or defective	Visual
114. Burr	Visual
115. Foreign matter	Visual

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4.3.2.1 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Groove width	Gage
202. Location of groove	Gage
203. Groove corner radii	Gage
204. Groove/thread angle	Gage
205. Length of cap	Gage
206. Boss diameter	Gage
207. Length of boss	Gage
208. Base thickness	Gage
209. Radii or chamfers missing or incorrect	Visual
210. Surface finish improper	Visual

4.3.2.2 Cap, End, Drawing C11740330.

CRITICAL	
None defined	
MAJOR	
101. Concentricity of groove to thread	Gage
102. Concentricity of thread pilot to thread	Gage
103. Concentricity of diameter to thread	Gage
104. Diameter of groove, maximum	Gage
105. Outer diameter, minimum	Gage
106. Pitch diameter of thread	Gage
107. Minor diameter of thread	Gage
108. Metal defective	Visual
109. Protective finish, inadequate or defective	Visual
110. Burr	Visual
111. Foreign matter	Visual
MINOR	
201. Groove width	Gage
202. Location of groove	Gage
203. Groove corner radii	Gage
204. Groove thread angle	Gage
205. Length of Cap	Gage
206. Diameter of thread pilot	Gage
207. Location of thread pilot	Gage
208. Chamfers missing or incorrect	Visual
209. Surface finish improper	Visual

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4.3.2.3 Motor Tube, Drawing D11740318.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Concentricity of inside diameter to thread	Gage
102. Concentricity of thread pilot to inside diameter	Gage
103. Concentricity of outside diameter to inside diameter	Gage
104. True position of nozzle hole diameter to inside diameter	Gage
105. Pitch diameter of threads	Gage
106. Minor diameter of threads	Gage
107. Thread pilot diameters, minimum	Gage
108. Inner diameter, maximum	Gage
109. Wall thickness, minimum	Gage
110. Metal defective	Visual
111. Protective finish, inadequate or defective	Visual
112. Burr	Visual
113. Foreign matter	Visual
MINOR	
201. Tube length	Gage
202. Hole location	Gage
203. Boss location	Gage
204. Boss width	Gage
205. Boss hole diameter	Gage
206. Bevel angle around hole	Gage
207. Bevel depth	Gage
208. Large outside diameter	Gage
209. Small outside diameter, minimum	Gage
210. Outside clearance diameter	Gage
211. Length, clearance diameter	Gage
212. Chamfers missing or incorrect	Visual
213. Boss blend radii	Visual
214. Width thread pilot	Gage
215. Thread lengths	Gage
216. Thread pilot chamfer diameter	Gage

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4.3.2.4 Tube, Motor Assembly, Drawing D11740308.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. True position of large thread to inside diameter	Gage
102. True position of small thread to large thread	Gage
103. Boss location dimension	Gage
104. Nozzle thread	Gage
105. Height of nozzle adapter	Gage
106. Trunnion angularity	Gage
107. Improper welding	Visual
108. Protective finish, inadequate or defective	Visual
109. Foreign matter	Visual
MINOR	
201. Trunnion location	Gage

4.3.2.5 Trunnion, Motor, Drawing B11740320.

CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of thread	Gage
102. Minor diameter of thread	Gage
103. Metal defective	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
201. Radius	Gage
202. Total length	Gage
203. Counterbore diameter	Gage
204. Counterbore depth	Gage
205. Outer diameter	Gage
206. Chamfer incorrect	Gage

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4.3.2.6 Trunnion, Motor, Drawing B11740334.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of thread	Gage
102. Minor diameter of thread	Gage
103. Metal defective	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
201. Radius incorrect	Gage
202. Total length	Gage
203. Counterbore diameter	Gage
204. Counterbore depth	Gage
205. Outer diameter	Gage
206. Chamfer incorrect	Gage

4.3.2.7 Adapter, Nozzle, Drawing B11740319.

CRITICAL	
None defined	
MAJOR	
101. Metal defective	Visual
102. Burr	Visual
103. Foreign matter	Visual
MINOR	
201. Total length	Gage
202. Radius incorrect	Gage
203. Outside diameter	Gage

4.3.2.8 Nozzle, Drawing B11740328.

CRITICAL	
None defined	

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4.3.2.8 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Concentricity of cone with thread	Gage
102. Concentricity of throat with thread	Gage
103. Throat diameter	Gage
104. Pitch diameter of threads	Gage
105. Major diameter of threads	Gage
106. Metal defective	Visual
107. Burr	Visual
108. Foreign matter	Visual
MINOR	
201. Length	Gage
202. Cone angle	Gage
203. Cone diameter	Gage
204. Chamfer width	Gage
205. Chamfer angle	Gage

4.3.2.9 Disk, Drawing B11740329.

CRITICAL	
None defined	
MAJOR	
101. Thickness	Gage
102. Metal defective	Visual
103. Burr	Visual
104. Foreign matter	Visual
105. Surface finish improper	Visual
MINOR	
201. Diameter	Gage

4.3.2.10 Spring Compression, Drawing B11740316.

CRITICAL	
None defined	

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4.3.2.10 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Load at specified heights	Gage
102. Solid height	Gage
103. Metal defective	Visual
104. Interference between coils at closed heights	Visual
105. Protective finish, inadequate or defective	Visual
106. Dry film, inadequate or defective	Visual
107. Burr	Visual
108. Foreign matter	Visual
MINOR	
201. Inner diameter, minimum	Gage
202. Outer diameter, maximum	Gage
203. Ends incorrect	Visual

4.3.2.11 Stop, Firing Pin, Drawing B11740317.

CRITICAL	
1. Thickness, overall	Gage
MAJOR	
101. Concentricity of shoulder to outside diameter	Gage
102. Concentricity of hole to outside diameter	Gage
103. Diameter of flange	Gage
104. Diameter, shoulder	Gage
105. Diameter, hole	Gage
106. Chamfer on hole	Visual
107. Metal defective	Visual
108. Burr	Visual
109. Foreign matter	Visual
110. Surface finish improper	Visual
111. Dry Film, inadequate or defective	Visual
MINOR	
201. Width of flange	Gage
202. Shoulder radius	Gage
203. Protective finish, inadequate or defective	Visual

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4.3.2.12 Firing Pin, Drawing C11740314.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Concentricity of diameter to striker	Gage
102. Perpendicularity of hole to slot	Gage
103. Diameter of striker	Gage
104. Length of striker	Gage
105. Radial profile of striker	Gage
106. Metal defective	Visual
107. Burr	Visual
108. Foreign matter	Visual
109. Surface finish, improper	Visual
110. Chamfers missing or incorrect	Visual
MINOR	
201. Diameter of flange	Gage
202. Flange width	Gage
203. Diameter, shank	Gage
204. Length to face of flange	Gage
205. Location of dowel pin hole	Gage
206. Diameter of dowel pin hole	Gage
207. Slot width	Gage
208. Location of slot	Gage
209. Slot depth	Gage
210. Width of flat	Gage

4.3.2.13 Housing, Firing Pin and Pellet Assembly, Drawing C11740313.

CRITICAL	
None defined	
MAJOR	
101. Concentricity of first groove on shank with shank diameter	Gage
102. Concentricity of shank with outside thread	Gage
103. Concentricity of small hole with outside thread	Gage
104. Concentricity of small thread with large thread	Gage
105. Concentricity of small thread undercut with outside thread	Gage
106. Concentricity of bore with thread	Gage
107. Pitch diameter of outside thread	Gage

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4.3.2.13 (Cont'd)

	Defect	Method of Inspection
MAJOR		
108.	Major diameter of outside thread	Gage
109.	Pitch diameter of inside thread	Gage
110.	Minor diameter of inside thread	Gage
111.	Diameter, shank	Gage
112.	Diameter of first groove from end on shank	Gage
113.	Width of first groove	Gage
114.	Groove diameter on outside threaded end	Gage
115.	Width of groove on outside threaded end	Gage
116.	Diameter of internal groove	Gage
117.	Width of internal groove	Gage
118.	Metal defective	Visual
119.	Burr	Visual
120.	Surface finish improper	Visual
121.	Foreign matter	Visual
122.	Radii or chambers missing or improper	Visual
123.	Pellet missing or improperly seated	Visual
124.	Protective finish inadequate or defective	Visual
MINOR		
201.	Total length	Gage
202.	Length to inside face of flange	Gage
203.	Length to outside face of flange	Gage
204.	Hex width	Gage
205.	Diameter of cavity	Gage
206.	Diameter of small hole	Gage
207.	Thickness of web at small hole	Gage
208.	Depth to inside undercut	Gage
209.	Thickness of flange, first groove	Gage
210.	Diameter of groove adjacent to flange	Gage
211.	Width of groove adjacent to flange	Gage
212.	Location of groove adjacent to flange	Gage
213.	Diameter of pellet hole	Gage
214.	Location of pellet hole	Gage
215.	Depth of pellet hole	Gage

4.3.2.14 Sear, Drawing C11740312.

CRITICAL

None defined

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4.3.2.14 (Cont'd)

	Defect	Method of Inspection
MAJOR		
101.	Concentricity of large groove with body diameter	Gage
102.	Concentricity of small groove with body diameter	Gage
103.	Symmetry of flat with body diameter	Gage
104.	Body diameter	Gage
105.	Diameter of hole	Gage
106.	Spherical radius in bottom of hole	Gage
107.	Slot width	Gage
108.	Location of slot	Gage
109.	Radius in bottom of slot	Gage
110.	Diameter of small groove	Gage
111.	Width of small groove	Gage
112.	Thickness of web at bottom of slot	Gage
113.	Thickness of web at bottom of hole	Gage
114.	Location of hole	Gage
115.	Metal defective	Visual
116.	Burr	Visual
117.	Surface finish improper	Visual
118.	Solid film, inadequate or defective	Visual
119.	Foreign matter	Visual
120.	Radii or chamfers missing or improper	Visual
MINOR		
201.	Length of part	Gage
202.	Diameter of large groove	Gage
203.	Radii of large groove	Gage
204.	Width of large groove	Gage
205.	Ramp thickness	Gage
206.	Ramp angle	Gage
207.	Ramp back angle	Gage
208.	Ramp radius	Gage
209.	Radius at base of ramp	Gage
210.	Location of ramp radius	Gage
211.	Length of ramp	Gage
212.	Location of spherical radius/slot	Gage
213.	Location of large groove	Gage
214.	Location of small groove	Gage
215.	Height of ramp	Gage
216.	Ramp end radius	Gage
217.	Depth of ramp radius	Gage

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4.3.2.15 Sear Housing, Drawing C11740311.

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Concentricity of long bore with thread	Gage
102.	True position of counterbore with long bore	Gage
103.	Major diameter of thread	Gage
104.	Pitch diameter of thread	Gage
105.	Diameter of shank	Gage
106.	Width of rectangular section	Gage
107.	Location of small cavity with rectangular face	Gage
108.	Total length	Gage
109.	Location of small cavity with end of shank	Gage
110.	Length of shank	Gage
111.	Length of shank to thread	Gage
112.	Diameter of long bore	Gage
113.	Diameter of counterbore	Gage
114.	Diameter of ball socket hole	Gage
115.	Location of ball socket hole	Gage
116.	Width of slot	Gage
117.	Location of slot	Gage
118.	Metal defective	Visual
119.	Burr	Visual
120.	Surface finish improper	Visual
121.	Foreign matter	Visual
122.	Radii or chamfers missing or improper	Visual
123.	Pellet missing or improperly seated	Visual
MINOR		
201.	Diameter of thread undercut	Gage
202.	Width of thread undercut	Gage
203.	Angle of thread relief	Gage
204.	Diameter of boss	Gage
205.	Diameter of small cavity	Gage
206.	Length of boss	Gage
207.	Pitch diameter of tapped holes	Gage
208.	Location of tapped holes	Gage
209.	Depth of counterbore	Gage
210.	Location of boss with center of large cavity	Gage
211.	Depth of small cavity	Gage
212.	Height from face of boss to flat	Gage
213.	Depth of long bore	Gage

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4.3.2.15 (Cont'd)

Defect	Method of Inspection
MINOR	
214. Diameter of pin hole, minimum	Gage
215. Location of pin hole from center of counterbore	Gage
216. Location of pin hole from center of large bore	Gage
217. Radius at base of small cavity	Gage
218. Diameter of pellet hole	Gage
219. Location of pellet hole	Gage
220. Depth of pellet hole	Gage
221. Chamfer at inner section of slot with hole	Visual

4.3.2.16 Pellet, Drawing B11740315.

CRITICAL	
None defined	
MAJOR	
101. Outer diameter	Gage
102. Length	Gage
MINOR	
None defined	

4.3.2.17 Nozzle Assembly, Drawing B11740327.

CRITICAL	
1. Any part missing	Visual
2. Improper assembly	Visual
MAJOR	
101. Diameter of internal cavity	Gage
102. Overall depth of cavity	Gage
103. Thread damaged	Visual
104. Rubber seal damaged	Visual
105. Disk damaged	Visual
106. Burr	Visual
107. Foreign matter	Visual
MINOR	
201. Angle	Gage
202. Depth of angle	Gage
203. Depth from base to angle	Gage

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4.3.2.18 Firing Pin Assembly, Drawing C11740310.

Defect	Method of Inspection
CRITICAL	
1. Any part missing	Visual
2. Improper assembly	Visual
MAJOR	
101. Threads damaged	Visual
102. Set screw sealant missing	Visual
103. Foreign matter	Visual
MINOR	
201. Exterior scratches	Visual

4.3.2.19 Rocket Motor, XM119, Metal Parts Assembly, Drawing D11740302.

CRITICAL	
None defined	
MAJOR	
101. Any part missing, except as otherwise classified	Visual
102. Improper assembly	Visual
MINOR	
None defined	

4.3.2.20 Box, Fiberboard (Inner), Drawing F9622824.

CRITICAL	
None defined	
MAJOR	
None defined	

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4.3.2.20 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Gap between body and cover, maximum	Manual
202. Contents loose	Manual
203. Ends loose or distorted	Manual/Visual
204. Cut or damaged through all impregnated layers	Visual
205. Marking missing, incorrect or illegible	Visual

4.3.2.21 Box, Packing, (Level A), Drawing C9622825.

CRITICAL

None defined

MAJOR

None defined

Defect	Method of Inspection
MINOR	
201. Contents loose	Manual
202. Strapping missing, loose, broken or improperly assembled	Manual/Visual
203. Contents of box exposed	Visual
204. Box damaged	Visual
205. Marking missing, incorrect, or illegible	Visual
206. Number of units incorrect	Visual

4.3.2.22 Box, Packing, (Level B), Drawing C9622826.

CRITICAL

None defined

MAJOR

None defined

Defect	Method of Inspection
MINOR	
201. Contents loose	Manual
202. Contents of box exposed	Visual
203. Ends loose or distorted	Visual/Manual
204. Cut or damaged through all layers	Visual
205. Marking missing, incorrect or illegible	Visual
206. Tape incomplete	Visual

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4.3.3 Inspection testing.-The following tests shall be performed in accordance with 4.4.

4.3.3.1 Hardness.-Fifty units from each heat-treated lot of components shall be tested for compliance with drawing requirement. Failure of any unit to comply with drawing requirements shall be cause for rejection of the lot.

4.3.3.2 Hydrostatic pressure.-Each component listed in 3.3 shall be subjected to an internal hydrostatic pressure. Failure of any component to comply with the requirements of 3.3 shall be cause for rejection of the component.

4.3.3.3 Firing pin release.-Each firing pin assembly shall be subjected to the firing pin release test. Failure of any assembly to comply with the requirements of 3.4 shall be cause for rejection of the assembly.

4.3.3.4 Magnetic Particle Inspection.-The outside surface of each motor tube shall be subjected to the magnetic particle inspection or penetrant inspection to determine the presence of cracks. Failure of any unit to comply with the requirements of MIL-I-6866 or MIL-I-6868, as applicable, shall be cause for rejection of the unit.

4.3.3.5 Protective finish.-The number of specimens and acceptance criteria for protective finish tests shall be in accordance with the applicable military standard listed on the drawing.

4.3.3.6 Residual Magnetism.-Each rocket motor assembly shall be tested for excessive residual magnetism. Failure of any unit to comply with 3.6 shall be cause for rejection of the rocket motor assembly.

4.3.4 Inspection equipment.-The examination and tests shall be made using equipment listed in LI 11740302.

4.4 Test Methods and Procedures.

4.4.1 Hardness.-Prior to the application of protective finish, a hardness testing machine shall be used to determine compliance with drawing requirements.

4.4.2 Hydrostatic pressure.-Prior to the application of protective finish, each component listed in 3.3 shall be individually mounted in a test device, using the fixture for sealing all openings, and subjected to the hydrostatic pressure to determine compliance with 3.3. The test device used shall internally seal the components at the prescribed seal surfaces without applying an axial load other than that applied by the liquid pressure.

4.4.3 Firing pin release.-The firing pin assembly shall be secured in an approved fixture. A pull shall be applied to the sear in the longitudinal direction in conjunction with the fixture which registers the amount of pull.

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4.4.4 Magnetic particle and penetrant.-The test shall be performed prior to the application of protective finish in accordance with the procedures established in MIL-I-6866 or MIL-I-6868 as applicable.

4.4.5 Protective finish.-The procedures for protective finish shall be in accordance with the applicable military standard referenced on the drawing.

4.4.6 Residual magnetism.-The assembly shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effect by placing the assembly 6 inches from and at the same height as the compass in a north-south horizontal position with the compass set in an east-west heading. The assembly shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 180 degrees in the horizontal position and moved slowly back past the compass.

5. PREPARATION FOR DELIVERY

5.1 Packing and Marking.-Packing and marking shall be as specified by procuring activity.

5.1.1 Level A.-Packing and marking shall be in accordance with Drawings F9266824 and C9266825.

5.1.2 Level B.-Packing and marking shall be in accordance with Drawings F9266824 and C9266826.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number, and date of this specification.

6.2 Mechanical failure.-A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

Custodian:

Army-MU

Preparing Activity:

Army - MU(FA)

Project No. 1377-A477

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

DIRECT GOVERNMENT CONTRACT ( ) SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE SPECIFICATION RESTRICTIVE?

( ) YES ( ) NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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